

Urban Challenges and Predictions – Data Driven Insights for Policy and Planning

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Abstract— The present paper illustrates a comparative analysis of 20 cities' urban sustainability activities in 2020 and 2023. Statistical and machine learning methods such as independent two-sample t-tests to compare notable differences between measures over time, Markov Chains to predict and simulate future traffic congestion states, and K-Means clustering to cluster cities by pollution and green area coverage are utilized. The results are presented in the form of an interactive Microsoft Power BI (Business Intelligence) dashboard and present actionable insights to urban planners. The results exhibit typical patterns of urban change and demonstrate the value of combining analytics and visualization for evidence-based policy making.

Keywords— Urban Sustainability, Markov Chains, K-Means Clustering, Statistics, Pollution, Smart Cities

I. INTRODUCTION

Cities have been growing steadily in both scale and activity, bringing with them a wide range of sustainability concerns. As urban areas accommodate more people, expand infrastructure, and drive economic output, the pressure on natural resources and public systems continues to build. According to the United Nations Department of Economic and Social Affairs (2018), approximately 68 percent of the global population is projected to live in urban areas by 2050. This shift is likely to result in higher pollution levels, worsening traffic, and a steady decline in green spaces, factors that can affect public health, liveability, and environmental balance in complex ways. Informed urban policy decisions require a data-driven understanding of the dynamic and multifaceted nature of urban development. Traditional policymaking, when grounded in outdated methods or incomplete data, often falls short in addressing complex challenges that vary across geographic regions and different types of urban environments. Thus, it is essential to compare urban indicators such as air pollution concentrations (e.g., PM_{2.5} concentration), traffic, population density, and vegetation cover using robust analytical tools.

II. PROBLEM STATEMENT

Today's cities are facing a mix of environmental and infrastructural issues such as deteriorating air quality, congested traffic conditions, shrinking green spaces, and increasing population densities. These issues are interlinked and threaten the liveability and sustainability of the city. As data becomes more and more available to gauge such problems, urban management practice can include exhaustive, comparative, and dynamic assessments that would help adaptive policy actions. There are also scarce available tools that enable the stakeholders to track change over time and test the effectiveness of initiatives. Cities are likely to undertake knee-jerk action if they are not offered a common analytical

platform for tracking, comparing, and interpreting urban sustainability indicators over time. This study addresses the need for an integrated approach to analysing urban sustainability by combining statistical methods, predictive modelling, and clustering techniques into an interactive visual platform. The goal is to empower decision-makers with clear, data-driven insights that can guide more sustainable urban planning.

The present research employs an exhaustive dataset of 20 Indian cities, comparing the most pivotal sustainability indicators of 2020 with those of 2023. Using statistical hypothesis testing, Markov Chain simulation, K-means clustering, and Power BI visualization, this study derives actionable insights into the evolution of urban metrics and identifies patterns across various urban typologies. The paper is intended to present a technical characterization of urban conditions and arm policymakers and urban planners with tools and insights to guide resource allocation and future urban development policy. Through integration of Python-based analysis and interactive Power BI dashboards, the research bridges the divide between advanced data analysis and user-friendly decision-support systems.

III. LITERATURE REVIEW

Early work on smart sustainable urbanism framed cities as complex, data-rich ecosystems that could be steered with advanced analytics. Bibri [1] argues that big-data science must sit at the heart of any future-oriented urban agenda, while his subsequent study with Krogstie [2] demonstrates how London and Barcelona translate that vision into practice through sensor grids, open data portals, and feedback-driven policy loops. Together, these studies establish the conceptual foundation for using pervasive data streams to pursue environmental and social goals.

Building on this foundation, Kandt et al [3] call for an "urban analytics for the long run," highlighting that short observational windows and one-off dashboards rarely influence durable policy. The research emphasize longitudinal analysis, reproducible methods, and the need for scalable platforms that evolve alongside the city and principles that guide the design of sustainable monitoring tools. At the methodological level, the study done by Yu et al in [4] shows that even simple Markov chains can predict short-term traffic flow with surprising accuracy, hinting at broader applications for state-transition models in urban forecasting. Ignaccolo et al [5] extend the discussion by applying functional clustering to air-quality networks, illustrating how grouping techniques reveal spatial patterns invisible to standard statistics and improving the strategic placement of sensors.

More recently, Chaturvedi and de Vries [6] catalogue machine-learning algorithms useful for land-use planning, underscoring the value of automated pattern recognition when zoning or redevelopment decisions outstrip human capacity to interpret data. Kaufmann et al [7] shows generative statistics that capture how amenities scale with population, yielding guidelines for equitable distribution of services as cities grow. The ecological stakes of urban growth are evident in Dubey and Kumar's [8] analysis of sprawl-induced environmental degradation, as well as [9] shows how pandemic lockdowns triggered abrupt improvements in air quality. Both papers demonstrate that urban conditions can shift rapidly—sometimes positively, sometimes negatively—reinforcing the need for monitoring systems that detect and interpret change in real time. The need for urban planning, flood mitigation strategies and water quality parameters were urged in [10,11]. Machine learning models and Markov models implemented in [12,13] provide good insight for the traffic congestion modelling and air quality index predictions.

Despite these advances, most studies isolate a single domain—traffic, air quality, land use—or apply one analytical lens at a time. Few combine statistical significance testing, clustering, and temporal modelling within a unified, interactive environment that stakeholders can query on demand. The present research responds to that gap by integrating T-tests, K-Means clustering, higher-order Markov chains, and live visualization into a single Power BI platform. This fusion not only quantifies past performance but also projects future trajectories, supplying policymakers with a dynamic, extensible toolkit for evidence-based urban sustainability planning.

IV. DATA COLLECTION AND PREPARATION

A. Data Collection

Seven key sustainability indicators are chosen: NO₂ levels (ppb), PM2.5 levels (µg/m³), Population density (people/km²), Traffic Congestion(%), Green Coverage(%), Public Transportation Usage(%) and Urban Expansion (%). These Indicators altogether cover environmental, demographic, and infrastructural spheres.

Data is collected from 20 Indian cities, with each city contributing 7 indicators for both 2020 and 2023, resulting in a dataset of 280 data points (20 cities × 7 metrics × 2 years). The cities were chosen to ensure diversity in geography, population, and urbanization scale. The year 2020 reflects pandemic-era values when urban activities are disrupted and by 2023, most cities resumed normal operations, offering a meaningful comparison to assess post-pandemic urban shifts. The data is collected from Government portals (Central Pollution Control Board, Smart Cities dashboard), Environmental databases (AQI), and Urban analytics platforms (World Population Review). Cross-verification across sources ensures consistency and reliability. In case of conflict, highest-confidence data (typically government) is used.

Cities are selected to represent various sizes, geographies, and urban typologies. This ensures application of clustering and comparison across heterogeneous conditions.

B. Data Preparation

Variables are normalized using Min-Max scaling presented in (1).

$$x' = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (1)$$

Where x is the original value (before scaling) and x' is the scaled value. This is applied independently to each feature x_i across all cities and both years. Cities with null values for any metric in either year are excluded.

Outliers are detected using Interquartile Range (IQR) as lower bound and upper bound expressed in (2) and (3):

$$\text{Lower Bound} = Q1 - (1.5 \times IQR) \quad (2)$$

$$\text{Upper Bound} = Q3 + (1.5 \times IQR) \quad (3)$$

Where $Q1$ represents the median of first quartile and $Q3$ represents the median of third quartile. Outliers are capped or excluded based on influence on clustering accuracy.

V. METHODOLOGY

A. T-test for Statistical Comparison

To verify if the differences between urban measurements for 2020 and 2023 are substantial, a paired T-test is performed. The paired T-test is employed to verify if two related groups' means is different, in this case, the numbers for a city in 2020 and 2023, in order to conclude if urban measurements have evolved over time.

In every urban indicator, a paired T-test is employed. The formula used for T-test is (4). The null hypothesis is there is no difference (5). The alternative hypothesis is that there is a significance difference (6).

$$t = \frac{d}{s_d / \sqrt{n}} \quad (4)$$

Where t is the test statistic, d is the mean of the differences between paired observations, s_d is the standard deviation of the differences, and n is the number of paired observations.

$$H_0: \mu_1 = \mu_2 \quad (5)$$

$$H_1: \mu_1 \neq \mu_2 \quad (6)$$

Significance level of $\alpha=0.05$ is taken. If p-value <0.05 , then the null hypothesis is rejected, and a significant difference between the years 2020 and 2023 is identified.

B. Markov Chain Model

Before applying the Markov Chain model, the initial step is to split the levels of traffic congestion into three separate states according to the Traffic congestion % values. The cities are split into the following states: High Congestion (>60%), Moderate Congestion (30-60%), and Low Congestion (<30%). These levels are selected to reflect average levels of city traffic congestion.

Markov Chains are used to model the transition probabilities among various states over time. In this research, the objective is to identify how the traffic congestion states of cities changed from 2020 to 2023.

The initial step is to compute the transition matrix. The transition matrix indicates the chance of transitioning from a state of congestion in 2020 to another in 2023. In other words, for every city, the 2020 congestion state is matched against 2023 congestion state, and transition probabilities for all the cities are calculated. The transition matrix would appear as given in (7). Every cell in the matrix is the transition probability between two congestion states.

$$M_a = \begin{bmatrix} P_{(H \rightarrow H)} & P_{(H \rightarrow M)} & P_{(H \rightarrow L)} \\ P_{(M \rightarrow H)} & P_{(M \rightarrow M)} & P_{(M \rightarrow L)} \\ P_{(L \rightarrow H)} & P_{(L \rightarrow M)} & P_{(L \rightarrow L)} \end{bmatrix} \quad (7)$$

Where, H represent High Congestion, M represent Moderate congestion and L represent Low congestion.

The odds that a city would be in the same state or shift the state are computed in the Markov matrix. The probability of a city's improvement or decline in traffic congestion is calculated. The cities are improving if they move from High to Low/Moderate congestion, worsened if they move from Moderate/Low to High congestion, or stay the same by remaining in the same congestion state.

C. K-means Clustering

In order to find patterns in the data, K -means clustering is used to cluster cities based on the urban statistics like pollution level, green coverage in 2023. The objective is to segment cities into a definite number of clusters, which have common traits when it comes to pollution and green cover.

Before utilization of K -means, every feature is standardized through StandardScaler using (8) so that all the measurements would have an equal contribution.

$$x_c' = \frac{x_i - \mu}{\sigma} \quad (8)$$

Where, x is the original value(before scaling), x_c' is the scaled value, μ is the mean and σ is the standard deviation of each feature.

K -means algorithm with $k=4$ is being used, where the number of clusters are chosen through visual inspection and domain expertise. The algorithm iteratively places every city in the nearest centroid and then updates the centroid by calculating average of all cities in cluster by using (9).

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (9)$$

The clusters are interpreted using the mean values of the urban indicators. For instance, clusters with high pollution are labelled "High Pollution"

D. Radar Chart

In order to present a graphical illustration of the urban features of every city, Radar charts or Spider plots are applied. The values graphed on the radar chart are PM2.5 levels, NO₂ levels, Traffic congestion, Green cover, and public transportation usage, which creates a balanced look into each city for the criteria being quality of life.

The features' values are normalized with MinMaxScaler (1) such that the measurements are on the same scale, and make the comparison of relative magnitudes easier graphically. For a chosen city, a radar chart is plotted by graphing urban measures' normalized values against axes that make up a polygon. There is one urban measure for every axis, and the city's value for the urban measure is given by the points on the chart. The form of the polygon gives a simple to visualize summary of an urban profile of a city.

E. Urban Ranking Index

Four urban indicators are chosen to create the Urban Ranking Index. The measurements are normalized to obtain comparable measurements between cities. The measurements

are normalized to the same range (0 to 1) to avoid the dominance of any one measure. The urban ranking of the city is established by the weighted formula (10) considering the factor of Quality of life.

$$\text{Urban Rank} = 0.15 \times \text{Population} - 0.25 \times \text{Traffic} + 0.25 \times \text{Public Transport} - 0.35 \times \text{PM2.5} \quad (10)$$

Here, *Population* reflects the city's population density, where higher values may indicate greater urbanization, *Traffic* represents the traffic congestion level, where higher values indicate more congestion, *Public Transport* represents the use of public transport, with higher values indicating greater usage, and *PM2.5* reflects the air quality level, with higher values indicating higher pollution levels.

The weights assigned: Population (0.15), Traffic (−0.25), Public Transport (0.25), and PM2.5 (−0.35), were selected based on their relative influence on urban livability, with their absolute values normalized to sum to 1. This ensures a balanced contribution across positive and negative factors. Higher negative weights were assigned to PM2.5 (−0.35) and Traffic (−0.25) due to their significant adverse impact on public health, air quality, and urban experience. Public Transport received a positive weight of (0.25), reflecting its importance in reducing emissions and congestion. Population was assigned a moderate positive weight (0.15) to capture urban scale and potential, while minimizing distortion from overpopulation. The resulting weighted formula captures a balanced, impact-driven Urban Rank.

For every city, the weighted formula of the normalized measures is computed using (10). The outcome is a composite urban ranking score that indicated the overall urban quality based on the chosen measures. Improved urban conditions are indicated by higher scores i.e. less traffic congestion, improved public transportation usage, and lower pollution.

VI. RESULTS AND DISCUSSION

A. T-test Analysis

T-test is employed to examine the difference in means of various urban indicators between 2020 and 2023. The research is interested in testing whether there existed a statistically significant difference between these indicators over the two periods.

As it can be observed in TABLE I, metrics like Traffic Congestion, Population Density, Urban Expansion and PM2.5 levels show significant difference (p-value < 0.05), indicating that these factors experienced substantial changes between 2020 and 2023. The metrics, such as NO₂ levels, Public Transport Usage and Green Coverage, did not exhibit significant change (p-value > 0.05), implying that these factors remained relatively stable over the period.

The results of the t-test are in line with the fact that population density, urbanization, and traffic congestion have increased from 2020 to 2023, and are correlated to each other as cities witness greater housing and infrastructure needs. PM2.5 reduction is an urgent city-level concern that must be met and stricter controls and environmental rules enacted. The fact that there is no overall change in NO₂ levels and the amount of green cover indicates reversals in achieving efficient progress in air quality and sustainable urbanization.

TABLE I. T-TEST SIGNIFICANCE TABLE

Metric	2020 Mean	2023 Mean	t-statistic	p-value	Significance
NO2 level	37.35	34.2	0.42	0.676	No
PM2.5 level	102.9	154.75	-2.6	0.0177	Yes
Population density (people/km ²)	13300.9	14372.3	-8.98	0.001	Yes
Traffic congestion (%)	33.15	37.8	-35.42	0.0062	Yes
Green Coverage (%)	20.67	20.27	0.66	0.518	No
Public Transport usage (%)	28.8	29.65	-1.27	0.22	No
Urban expansion (km ²)	544	554.5	-3.59	0.0019	Yes

B. Markov Chain Analysis

Markov Chain model is being used to model the transition of the traffic congestion states of the cities from 2020 to 2023. It has been determined through this modelling how the cities transition from High, Moderate, and Low congestion states to three congestion states.

Transition probabilities as seen in TABLE II indicates that most cities declined. No city has gone from worse traffic congestion to better. Cities which experienced high-speed urbanization and poor public transport are more congested.

TABLE II. MARKOV TRANSITION PROBABILITIES

State	High	Low	Medium	Total
High	1.00	-	-	1.00
Low	-	0.63	0.37	1.00
Medium	0.09	-	0.91	1.00
Total	1.09	0.63	1.28	3.00

Markov Chain analysis validates the intuition that traffic congestion is a dynamic parameter and that cities either improve or deteriorate depending on urban planning policy. The results show that concerted efforts in public transport and traffic management development can reduce congestion levels in the long term.

C. K-means Clustering

K-means clustering analysis separated cities into four categories according to urban indicators Pm2.5 concentration and green coverage. Cluster model revealed diverse urbanization and sustainability patterns at the city scale for cities. The clusters revealed four distinct types of cities: High Pollution, Low green coverage, Low Pollution and High green coverage, and Moderate Pollution, Balanced Metrics.

The distribution of cities across the clusters highlighted in Fig. 1 show that most cities are concentrated in the High Pollution cluster, while a few cities exhibited moderate values.

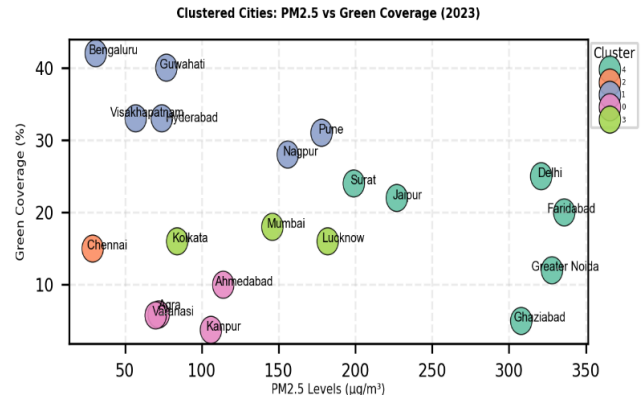


Fig. 1. City clusters formed by K-means clustering

The clustering indicates the disparity in urban planning outcomes. Urban areas that are dense with pollution and lacking in greenery are particularly hard to make liveable. Urban areas that are dense with greenery and low with pollution are instances of good urban planning because they are mindful of sustainability

D. Radar Chart

Radar charts are used to represent cities' urban profiles, graphing key indicators and contrast the overall urban environment and sustainability of cities in 2020 and 2023. The radar charts diagrammatically depicted the dissimilar urban traits of cities as seen in Fig. 2 and Fig. 3. Some cities possess good public transport and adequate greenery, whereas others experience high congestion and poor air.

More evenly balanced profiles, with increased use of public transport, less traffic congestion, and lower levels of pollution are also correlated with placing cities higher on the radar charts.

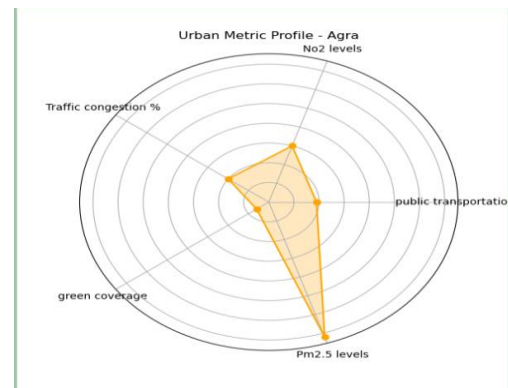


Fig. 2. Radar Chart of Agra

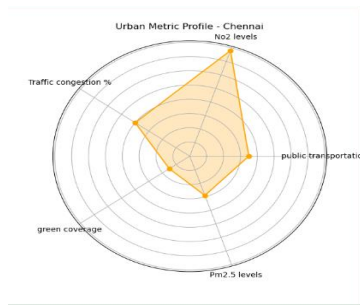


Fig. 3. Radar Chart of Chennai

The radar chart visualization provides a simple way of establishing the trade-offs and the relative strength of each city. The charts indicates that certain cities are strong on public transport and green space but lacking in traffic congestion and pollution, with implications for more integrated urban policy.

E. Overall Insights and Policy Implications

The combination of t-tests, K-means clustering, Radar charts, and Markov Chain analysis provides a comprehensive view of the trends of urban development of cities between 2020 and 2023.

The findings show that reduced traffic congestion, better air quality, and greater use of public transport are making cities the urban pioneers. Policy makers can give priority to these factors to ensure sustainable urban life. Dramatic decreases in Pm2.5 levels and better use of public transport indicates intensified efforts to mitigate pollution and encourage sustainable mobility.

Presently, there is a case of urban inequality, where certain cities are placed well in terms of sustainability, but others are confronted with congestion and pollution. These variations highlight the necessity for interventions that are tailored to the specific urban issues of each city

All of these insights are gained from the intensive 6 page Power BI dashboard that displays in depth analysis of all 20 cities for both 2020 and 2023 as seen in Figs. 4–9. These dashboards are interactive and easy to use.

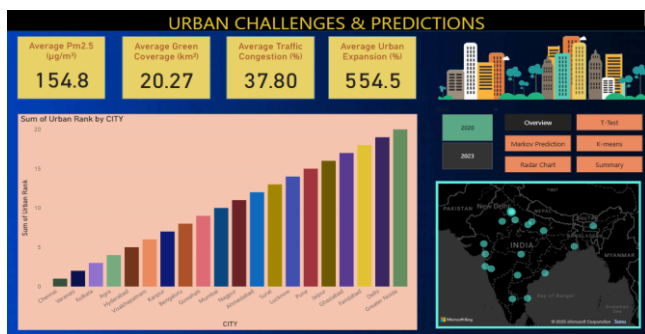


Fig. 4. Overview Page

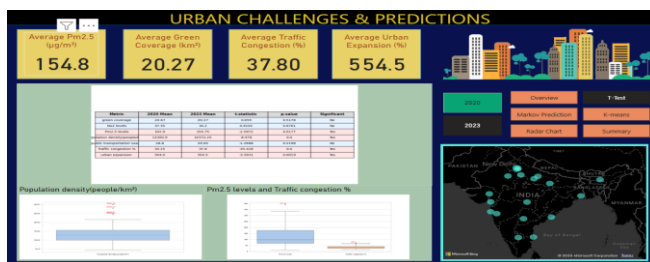


Fig. 5. T-test Dashboard

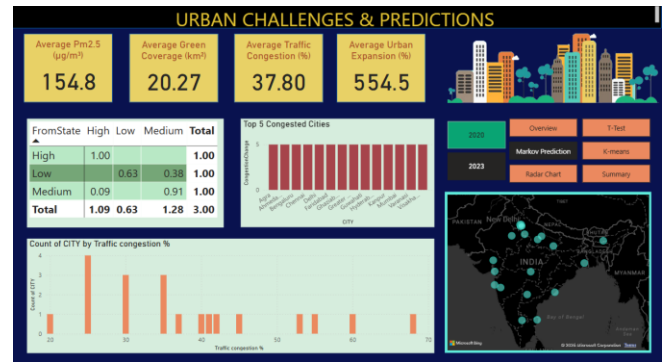


Fig. 6. Markov Chain Dashboard

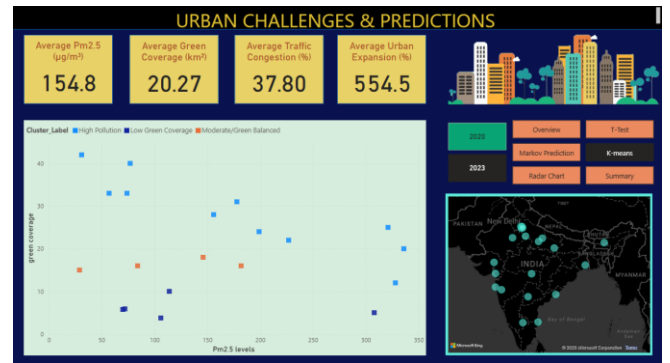


Fig. 7. K-means Clustering Dashboard

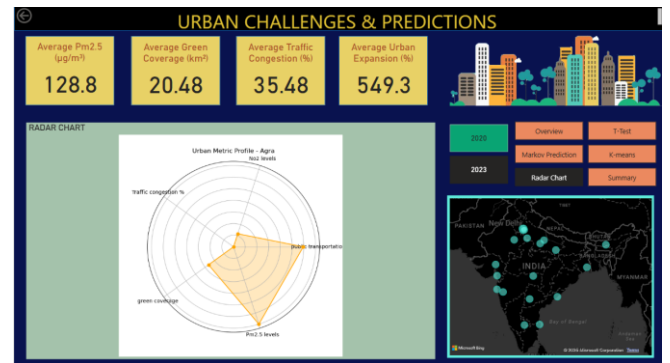


Fig. 8. Radar Chart Dashboard

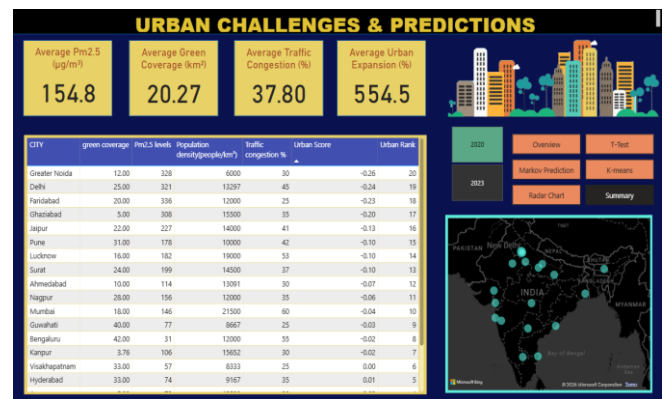


Fig. 9. Summary Dashboard

VII. CONCLUSION

The research provides an integrated, data-based evaluation of city sustainability across twenty cities by contrasting primary indicators for 2020 and 2023. The study is done through a multi-model pipeline that combines statistical inference, unsupervised learning, probabilistic modelling, and interactive visualization using Power BI.

K-Means clustering is employed as a specific technique to group cities based on sustainability behaviour. The clusters are semantically labelled: Cluster 0 for "Low Green Coverage", Cluster 1 for "Low Pollution, High Green Coverage", Cluster 2 for "Moderate/Balanced" cities, and Cluster 3 for "High Pollution" cities. This semantic interpretation made the clusters qualitative labels for describing states of urban sustainability.

To convey how cities evolved from year to year in states of traffic congestion, a Markov chain transition model is constructed from the cluster assignments for each year. This provided transition probabilities describing whether a city is probably going to improve, deteriorate, or remain the same in sustainability performance—thus transforming static clustering into a dynamic prediction tool. Paired T-tests are conducted for each of the seven measures to check statistical significance of change that is witnessed between 2020 and 2023. Along with measuring improvement or decline, the tests also ensured trend reliability as reported by scoring and clustering.

Finally, the entire pipeline is unified into a comprehensive, interactive Power BI dashboard. The dashboard allowed users to explore clustering, statistical significance, Markov transitions, spider/radar plots, and an urban score index constructed manually, giving a single window into the status and journey of each city.

While the current model offers valuable insights, its potential can be greatly expanded by extending the dataset to cover additional years. Doing so would enable time-series analysis and the implementation of higher-order Markov chains, allowing for a deeper understanding of how urban sustainability indicators evolve over time. Moreover, the inclusion of broader metrics such as water usage, waste generation, and energy efficiency would enrich the analysis, offering a more holistic view of a city's sustainability profile.

In summary, the paper presents a comprehensive and innovative approach to analyzing urban sustainability by bringing together statistical testing, machine learning techniques, and interactive visualization. Instead of relying solely on basic summaries or static data, it uses clustering to identify patterns among cities, statistical tests to measure progress over time, and forecasting models to anticipate future sustainability trends. All these elements are seamlessly integrated into a Power BI dashboard, making complex analyses accessible and visually engaging. The inclusion of tools like interactive radar charts and a simplified Urban Score Index allows for intuitive comparison and interpretation, enabling both policymakers and the public to better understand urban performance. By offering a scalable and adaptable model, the work can set a foundation for more informed, data-driven decision-making and long-term sustainability planning across cities.

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